

```
-- set operators
-- In MySQL set operators are used to combine the results of two or more select
-- queries into a single result.
-- Example of set operators in SQL
-- 1) Union 2) Union all 3) Intersect 4) Minus(Except)
```

```
-- rules of set operators:-
-- Number of columns must be the same.
-- Data type must be compatible
-- Column names in results come from the first query.
-- order by can be used but in the end of the query.
```

```
-- 1) UNION:-
-- Union combines results and removes duplicates
-- Syntax
```

```
/* Select column1 int, column2 varchar(225)
from table1
union
select clolumn2 int, column2 varchar(225)
from table2;
*/
```

```
/*
Select column1 int, column2 varchar(225)
from table1
union
select clolumn2 int, column2 varchar(225)
from table2
union
select clolumn2 int, column2 varchar(225)
from table3;
*/
```

```
create database if not exists setOperators;
use setOperators;
create table employees(emp_id int primary key auto_increment,
emp_name varchar(225) not null, email varchar(225),
emp_phone_no varchar(12)) auto_increment = 101;
```

```
create table managers(manager_id int primary key auto_increment,
manager_name varchar(225) not null, email varchar(225),
manager_phone_no varchar(12)) auto_increment = 10001;
```

```
INSERT INTO employees(emp_name,email,emp_phone_no) VALUES
('Amit Sharma','amit@company.com','9876500001'),
('Neha Verma','neha@company.com','9876500002'),
('Rahul Mehta','rahul@company.com','9876500003'),
('Priya Kapoor','priya@company.com','9876500004'),
('Vikas Singh','vikas@company.com','9876500005'),
('Anjali Gupta','anjali@company.com','9876500006'),
```

```
( 'Rohit Jain', 'rohit@company.com', '9876500007'),
( 'Sneha Iyer', 'sneha@company.com', '9876500008'),
( 'Karan Malhotra', 'karan@company.com', '9876500009'),
( 'Meera Joshi', 'meera@company.com', '9876500010'),
( 'Arjun Nair', 'arjun@company.com', '9876500011'),
( 'Simran Kaur', 'simran@company.com', '9876500012'),
( 'Aditya Roy', 'aditya@company.com', '9876500013'),
( 'Pooja Bansal', 'pooja@company.com', '9876500014'),
( 'Varun Khanna', 'varun@company.com', '9876500015'),
( 'Nikhil Arora', 'nikhil@company.com', '9876500016'),
( 'Ritika Shah', 'ritika@company.com', '9876500017'),
( 'Siddharth Rao', 'sid@company.com', '9876500018'),
( 'Tanya Desai', 'tanya@company.com', '9876500019'),
( 'Dev Patel', 'dev@company.com', '9876500020');
```

```
INSERT INTO managers(manager_name,email,manager_phone_no) VALUES
( 'Amit Sharma', 'amit@company.com', '9876500001'),
( 'Priya Kapoor', 'priya@company.com', '9876500004'),
( 'Rohit Jain', 'rohit@company.com', '9876500007'),
( 'Simran Kaur', 'simran@company.com', '9876500012'),
( 'Varun Khanna', 'varun@company.com', '9876500015'),
```

```
( 'Manish Tiwari', 'manish@company.com', '9876600001'),
( 'Kavita Sethi', 'kavita@company.com', '9876600002'),
( 'Raj Malhotra', 'raj@company.com', '9876600003'),
( 'Sunita Yadav', 'sunita@company.com', '9876600004'),
( 'Deepak Chauhan', 'deepak@company.com', '9876600005');
```

```
INSERT INTO managers(manager_name,email,manager_phone_no) VALUES
( 'Aakash Mehra', 'aakash@company.com', '9876700001'),
( 'Ritu Aggarwal', 'ritu@company.com', '9876700002'),
( 'Mohit Bhatia', 'mohit@company.com', '9876700003'),
( 'Shalini Tripathi', 'shalini@company.com', '9876700004'),
( 'Tarun Saxena', 'tarun@company.com', '9876700005');
```

```
select * from employees;
select * from managers;
```

```
select emp_id,emp_name as all_employees_name from employees
union
select manager_id,manager_name as management_name from managers
order by all_employees_name;
```

```
-- 2) Union all
-- Union all combines results but keeps duplicates.
CREATE TABLE students(
student_id INT PRIMARY KEY AUTO_INCREMENT,
student_name VARCHAR(100) NOT NULL,
```

```

email VARCHAR(100)
) AUTO_INCREMENT = 101;
INSERT INTO students(student_name,email) VALUES
('Amit Sharma','amit@gmail.com'),
('Neha Verma','neha@gmail.com'),
('Rahul Mehta','rahul@gmail.com'),
('Priya Kapoor','priya@gmail.com'),
('Vikas Singh','vikas@gmail.com'),
('Anjali Gupta','anjali@gmail.com'),
('Rohit Jain','rohit@gmail.com'),
('Sneha Iyer','sneha@gmail.com'),
('Karan Malhotra','karan@gmail.com'),
('Meera Joshi','meera@gmail.com'),
('Arjun Nair','arjun@gmail.com'),
('Simran Kaur','simran@gmail.com'),
('Aditya Roy','aditya@gmail.com'),
('Pooja Bansal','pooja@gmail.com'),
('Varun Khanna','varun@gmail.com'),
('Nikhil Arora','nikhil@gmail.com'),
('Ritika Shah','ritika@gmail.com'),
('Siddharth Rao','sid@gmail.com'),
('Tanya Desai','tanya@gmail.com'),
('Dev Patel','dev@gmail.com');

```

```

CREATE TABLE cricket_team(
player_id INT PRIMARY KEY AUTO_INCREMENT,
player_name VARCHAR(100) NOT NULL
);

```

```

INSERT INTO cricket_team(player_name) VALUES
('Amit Sharma'),
('Rahul Mehta'),
('Priya Kapoor'),
('Rohit Jain'),
('Sneha Iyer'),
('Arjun Nair'),
('Simran Kaur'),
('Varun Khanna'),
('Nikhil Arora'),
('Dev Patel');

```

```

CREATE TABLE football_team(
player_id INT PRIMARY KEY AUTO_INCREMENT,
player_name VARCHAR(100) NOT NULL
);

```

```

INSERT INTO football_team(player_name) VALUES
('Rahul Mehta'),
('Priya Kapoor'),
('Sneha Iyer'),

```

```
('Arjun Nair'),  
('Simran Kaur'),  
('Varun Khanna'),  
('Dev Patel'),  
('Neha Verma'),  
('Vikas Singh'),  
('Anjali Gupta'),  
('Karan Malhotra'),  
('Meera Joshi'),  
('Aditya Roy'),  
('Pooja Bansal'),  
('Ritika Shah');
```

```
select * from students;  
select * from cricket_team;  
select * from football_team;
```

```
-- find students who play either cricket or football.  
select player_name from cricket_team  
union  
select player_name from football_team;  
-- find students who play either cricket or football with duplicates.  
select player_name from cricket_team  
union all  
select player_name from football_team;
```

```
-- Intersect :-  
-- Intersect returns matching records/overlapping records from both table  
select player_name from cricket_team  
intersect  
select player_name from football_team;
```

```
-- Except:-  
-- Except returns unique records from 1st table.  
select player_name from cricket_team  
except  
select player_name from football_team;
```

```
-- student who don't play any sports  
select * from students;  
select * from cricket_team;  
select * from football_team;
```

```
select student_name from students
except
(select player_name from cricket_team
union all
select player_name from football_team);
```

```
-- student who play both sports and are in student table
-- players in teams but not in student table
-- student who play exactly one sport
```

```
select player_name from football_team
except
select player_name from cricket_team
;
```